■# PAPER<i>284: M16 Eagle Nebula UQFF — Dual Mass Co-Action Product (Φ</i>dm)

Author: Daniel T. Murphy | **Date:** March 2026

__Session:__ 80 | __Module:__ M16UQFFMODULE.cpp (22nd C++ UQFF module)

■# PAPER*284: M16 Eagle Nebula UQFF — Dual Mass Co-Action Product (*Φdm)

SFR Growth × Photoevaporation Erosion Multiplicative Coupling

Classification: UQFF 2.0 Gravitational Physics — Nebular Mass Dynamics **System:** M16 Eagle Nebula (IC 4703), Eagle Nebula Star-Forming Region **Session:** 80 | **Module:** M16*UQFF*MODULE.cpp (22nd C++ UQFF module) **Author:** Daniel T. Murphy | **Date:** March 2026

Abstract

This paper introduces the **Dual Mass Co-Action Product** (Φ*dm) — a UQFF gravity modulation factor that couples star-formation-driven mass accumulation and radiation-driven photoevaporation erosion through a* ***multiplicative*** *product rather than the previously used additive form. For M16 (Eagle Nebula), with star formation rate SFR = 1 M■/yr over initial gas mass M■ = 1200 M■, and maximum photoevaporation fraction E■ = 0.3 (30%), the multiplicative form produces a 24.3% reduction in* Φdm relative to the additive approximation at t = 5 Myr. This is the **first UQFF module** to simultaneously apply an additive-gain and saturation-subtractive product on the same gravity term.

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2. Physical Motivation

In active star-forming nebulae, two competing processes drive mass evolution:

- **Star Formation Accretion** — molecular gas accretes onto protostars, increasing the effective gravitational mass fraction by SFR_rate × t.
- **Photoevaporation Erosion** — UV radiation from newly formed massive stars erodes the surrounding gas, progressively reducing the effective mass by a saturating fraction E_rad(t).

The **additive approximation** (used in prior UQFF modules):

```
\Phi_{dm}^{add} = g_{base} \times (1 + M_{sf}) - E_{rad}
```

linearly superposes the two effects, implicitly treating them as independent processes acting on separate mass reservoirs.

The **multiplicative form** (this paper):

```
\Phi_{dm}(t) = (1 + \text{SFR\_rate} \times t) \times (1 - E_{rad}(t))
```

correctly encodes that **the eroded mass is drawn from the same growing reservoir** — the fraction lost to photoevaporation scales with the mass being accreted, not the original quiescent mass. This is physically accurate for pillar-geometry star formation (e.g., M16's "Pillars of Creation").

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3. Mathematical Formulation

3.1 Parameters (M16 Eagle Nebula)

Parameter	Value	Description
M■	2.387×10^{33} kg (1200 M■)	Initial nebula gas mass
SFR	1 M■/yr	Star formation rate
SFR_rate	2.639×10^{-11} s^{-1}	= SFR / (M■/M■) / (3.156×10■ s/yr)
τ_erode	9.468×10^{13} s (3 Myr)	Photoevaporation e-folding time
E■	0.3	Maximum photoevaporation fraction
g_base	1.454×10^{-12} m/s^2	G × M / r^2 at r = 3.31 × 10^{1}■ m

3.2 Dual Co-Action Product

```
M_{sf}(t) = \text{SFR\_rate} \times t
```

```
E_{rad}(t) = E_0 \left(1 - e^{-t/\tau}\right)
```

```
\boxed{\Phi_{dm}(t) = (1 + M_{sf}) \times (1 - E_{rad})}
```

3.3 Dynamic Gravity Term

```
g_{dyn}(t) = g_{base} \times \Phi_{dm}(t)
```

3.4 Multiplicative–Additive Gap

The gap relative to the additive form:

```
\Delta_{gap} = \Phi_{dm}^{mult} - \Phi_{dm}^{add} = -(M_{sf} \times E_{rad})
```

This cross-term is always **negative** — the multiplicative form predicts lower gravity than the additive approximation whenever both SFR accumulation and erosion are simultaneously active.

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4. Numerical Results at t = 5 Myr

t = 5 Myr = 1.578 × 10^{1}■ s

Quantity	Value
M*sf*frac	SFR_rate × t = 2.639×10^{-11} × 1.578×10^{1}■ = **4164.8**
E_rad	E■ × (1 – exp(–5/3)) = 0.3 × 0.8110 = **0.2433**
Φ_dm (multiplicative)	(1 + 4164.8) × (1 – 0.2433) = 4165.8 × 0.7567 = **3151.9**
Φ_dm (additive)	(1 + 4164.8) – 0.2433 = **4165.6**
gap*mult*add	–(4164.8 × 0.2433) = **–1013.3** (24.3% less)
g_dyn(5 Myr)	1.454×10^{-12} × 3151.9 = **4.583 × 10⁻■ m/s^2**

The multiplicative gap of –1013.3 confirms that treating erosion as acting on the growing mass reservoir (not the static initial mass) produces a **measurable 24.3% reduction** compared to the additive approximation.

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5. Connection to UQFF 2.0 g_total

In the full M16 UQFF 2.0 equation:

```
g_{total}(r, t) = \left[g_{dyn}(t) + U_{g,sum}(26) + \Lambda + Q + L + F + g_{exp}\right]
\times \text{corr}_{SC}
```

The Φ*dm product modulates only the dynamic base gravity term* gdyn. The 26-layer Triadic (U*g,sum), cosmological* Λ*, quantum, Lorentz, fluid, and Friedmann expansion terms are all independent of* Φdm — the modulation is cleanly scoped to the time-evolving mass component.

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6. UQFF Historical Distinction

Module	SFR Term	Erosion Term	Form
Session 55 CP3 M16EagleNebulaRadiationSFR	g*base×(1+M*sf)	−E_rad	**Additive**
This paper (PAPER_284)	(1+M_sf)	×(1−E_rad)	**Multiplicative**

This is the **first UQFF module** to use the multiplicative dual co-action form, correctly encoding the coupled feedback between star formation accretion and pillar photoevaporation for M16-class nebulae.

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7. Wolfram KB Term

```
M16UQFF:Phi_dm=(1+SFR_rate*t)*(1-E_0*(1-Exp[-t/tau])); SFR_rate=2.639e-11/s;
M_sf_frac=SFR_rate*t [PAPER_284]
```

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8. Cross-References

PAPER_285: Erosion Saturation Half-Time (t_half, ΔgMax)

PAPER_286: Nebular Friedmann Redshift (κ_neb, z=0.0015)

M16*UQFF*MODULE.cpp: Full UQFF 2.0 C++ implementation (22nd module)

CondensedPhysics3.py: `M16DualMassCoActionProductCalculator`
